PROGRAMMING FUNDAMENTALS

Lab Report 01 — C++ Basics & Environment Setup

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Semester / Sec:	1st Semester (Section C)	**Instructor:**	Sir Asad Sahib
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TASK 01: Hello World Program

Basic Output Stream

Description: Writing and compiling the basic boilerplate code in C++ to display a string output on the screen using the `std::cout` stream.

```
#include <iostream> using namespace std; int main() { cout << "hello world" << endl; return 0; }
```

CONSOLE OUTPUT

```
hello world

[process completed - press Enter]
```

TASK 02: Multi-line Text Output

String Formatting & Newlines

Description: Printing multiple formatted text statements onto separate lines using `std::endl`.

```
#include <iostream> using namespace std; int main() { cout << "You are 10 years old" << endl; cout << "You are too young to play the game." << endl; return 0; }
```

CONSOLE OUTPUT

```
You are 10 years old
You are too young to play the game.

[process completed - press Enter]
```

TASK 03: Printing Asterisk Pattern

Console Graphics / ASCII Art

Description: Generating a structured right-angled triangle pattern of asterisks using consecutive output statements.

```
#include <iostream> using namespace std; int main() { cout << "*" << endl; cout << "**" << endl; cout << "***" << endl; cout << "****" << endl; cout << "*****" << endl; return 0; }
```

CONSOLE OUTPUT

```
*
**
***
****
*****

[process completed - press Enter]
```